

Healthy Me: a personal visual health journal for self-reflection and knowledge sharing

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Wearable sensors and personal digital health tracking devices capture just the physiological metrics and fail to include perceptions of emotional and social wellbeing metrics to provide a personalized and actionable care plan. We introduce

"Healthy Me", a personal visual health data journaling application to explore whether the combination of **subjective measures** and the **quantified self-technologies** can help a person develop their own digital wellbeing stories as well as enhancing the dialogue between health providers. We carried out a qualitative study with both patient participants and healthcare professionals.

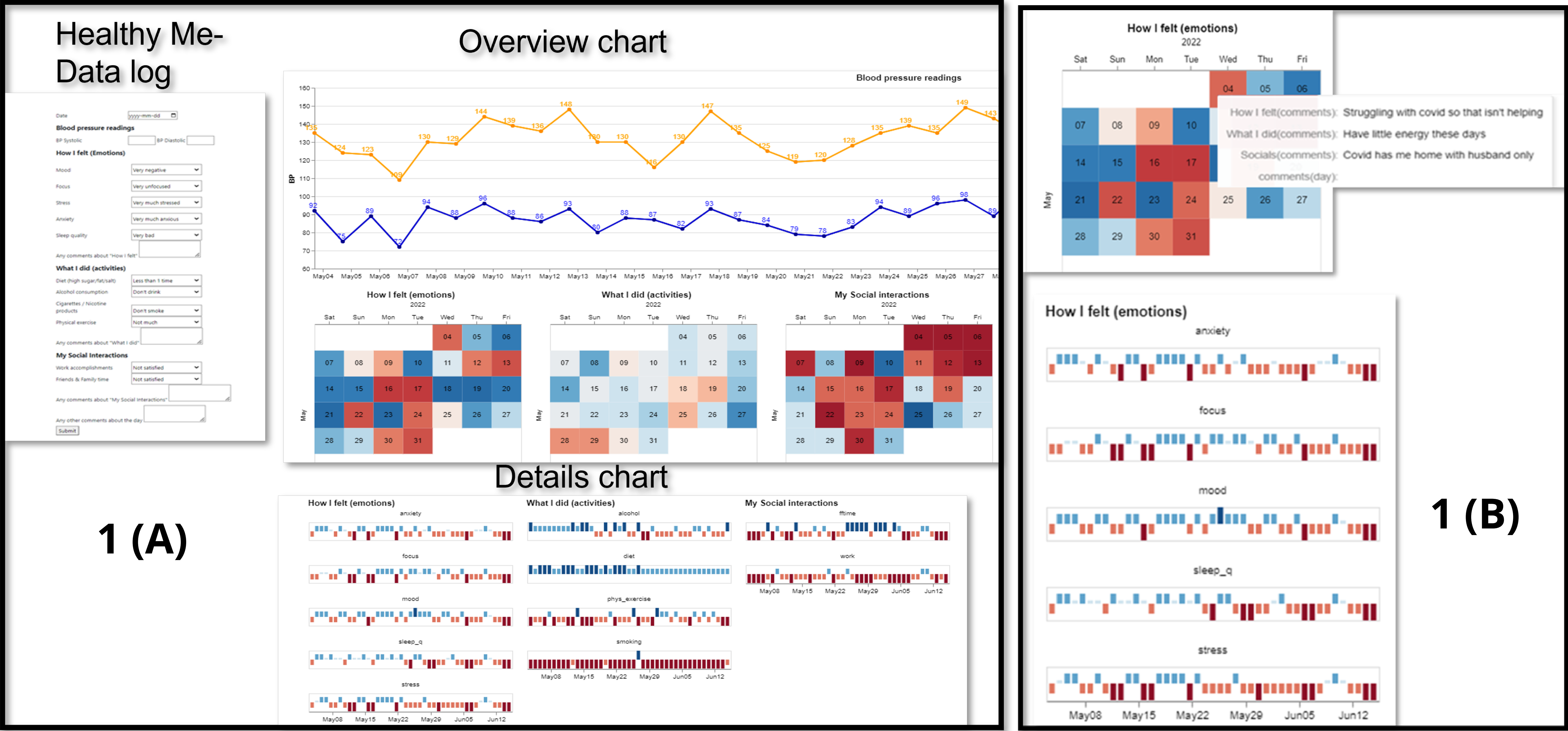


Figure 1(A):Healthy Me Design Interface: data log form(left), overview with calendar charts (center), detail view with diverging bar charts (bottom) , Figure 1(B): How I felt metrics with patient participant's journal entry

Preliminary results from the qualitative study

Themes	Description	Results	Participants feedback
Bringing in subjective metrics	tracking subjective metrics like emotional, physical and social behaviors	Self-awareness Invoked curiosity	P1: "I think it makes me more aware of sort of my lifestyle, my, my moods, what I'm doing, how I'm eating? I don't like seeing those 140 and above numbers, you know, I want to see something, between 120 and 130? And if I'm, not in there, then what can I do to get this better?" P3: "Like for me, I would be curious to know if my blood pressure is in direct relation to my migraines, for example."
Data interpretation	making sense of what is being displayed in the visualization	Pattern recognition	P1: "if you have the blood pressure on a line graph, and then you have the moods and stuff, you go, hey, I can see some, you know, correlations between these two here, I was, this was not a good day, my blood pressure was up a bit., my mood was calm, relaxed, and yeah, it was fine."
Enhancing communication	communication between health providers and patients	Quick snapshot	P2: "I would certainly share it with an internist, and he wants a snapshot of my health over the last year or so. So, I think I'll share a copy of this with him."
Engagement	continuity of self-tracking	Proactive	P1: "I have to be more proactive about my health. And yeah, this keeping data is a really good way. I'm definitely going to do that. Continue to choose it because I think it will just solidify it as a habit."

Themes	Results	Health Professionals feedback
Current Challenges	Patient literacy and motivation	HP2: "And the biggest obstacle we had was people using their computer. These were generally older people".
Usability of "Healthy Me" application	Educational tool	HP1: "I think it can be educational for the patient, or maybe help them gain insight into how maybe, you know, stress, or, alternatively, positive social interactions might be affecting their aspects of health. So I think it could be a really good teaching tool, but also help with self awareness."
Making sense of data through visualization	Quick snapshot	HP3: "And it kind of is nice, you can track over the last week or the last month and kind of see like those ebbs and flows versus you're just going like, a table with a bunch of numbers."
Clinical decision-making	Mixed reviews	HP1: "as a provider, it's also very helpful. If someone's doing especially like serious blood pressure monitoring, then I can also put that in my EMR, and it's easier for me then to decide if I need to titrate medication or do more teaching on health behaviors." HP2: "Well, I guess one of them is accuracy. reliability. There's probably still a fair amount of skepticism about patient entered data."

Future work : Mobile compatibility

Although "Healthy Me" application was designed for larger screen like laptops and tablets, participants used the application on mobile phones because of convenience. While this also meant we saw very little use of brushing interactions, we also saw the importance of the "notebook in the pocket" model of a personal health journal.

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